Ibrahim Odat

AI Security Researcher · LLM-Enabled Autonomous Systems Security

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SUMMARY

AI security researcher specializing in the security of large language model (LLM)-enabled autonomous drone systems and multi-agent unmanned aerial vehicle (UAV) architectures. Holds an M.S. in Cybersecurity (GPA 4.0/4.0) from Oakland University and is pursuing a Ph.D. in Computer Science & Informatics under Dr. Anyi Liu, with research addressing emerging vulnerabilities in AI-driven autonomous systems with implications for national defense, border surveillance, and critical infrastructure protection. Author of three peer-reviewed IEEE and IEEE/ACM publications and one preprint, cited 52 times (Google Scholar, November 2026). Current doctoral research identifies memory as a control-plane attack surface in LLM-based multi-agent systems, with a defense pipeline that reduces measured attack success from 100% to 0% in controlled evaluation.

EDUCATION

Oakland University — Rochester, MI
Ph.D. Candidate, Computer Science & Informatics · GPA 3.93/4.0 — Jan. 2025 – Expected Jan. 2029

- Dissertation (in progress): Security of LLM-Enabled Multi-Agent UAV Systems
- Advisor: Dr. Anyi Liu, Department of Computer Science & Engineering
- Research focus: adversarial threats, memory-poisoning attacks, and defense mechanisms for LLM-powered autonomous drone and multi-agent systems

Oakland University — Rochester, MI
M.S., Cybersecurity · GPA 4.0/4.0 — May 2022 – Dec. 2023

PUBLICATIONS

Total citations: 52 (Google Scholar, November 2026)

When Memory Hurts: Amplifying Mission-Channel Attacks in LLM-Driven UAV Agents — 2026
25th International Conference on Machine Learning and Applications (ICMLA 2026), pp. 693–700

- Authors: **Odat, I.**, Tawayha, H., & Liu, A. (First Author) · doi:10.1109/ICMLA69890.2026.00101
- Measured whether bounded short-term mission memory amplifies one-shot corruptions of the evidence a UAV agent sees, across four attack pathways, 12 planners, and 1,920 runs in a PX4/MAVSDK-compatible mission framework
- Showed that bounded memory raises attack-side mission failure from 31.5% to 53.3%, with false perception amplified 2.83×; a write-time Memory Contract Monitor reduced belief persistence to zero under direct memory poisoning

Net-GPT: A LLM-Empowered Man-in-the-Middle Chatbot for Unmanned Aerial Vehicle — 2023
IEEE/ACM Symposium on Edge Computing (SEC 2023), pp. 287–293

- Authors: Piggott, B., Patil, S., Feng, G., **Odat, I.**, Mukherjee, R., Dharmalingam, B., & Liu, A. · doi:10.1145/3583740.3626809
- Contributed core LLM framework design, system implementation, and experimental evaluation of the attack pipeline; developed and validated the LLM fine-tuning methodology that achieved 95.3% accuracy in generating counterfeit UAV network packets, enabling automated man-in-the-middle (MITM) attacks on drone-to-ground-station communications
- Cited 50 times on Google Scholar as of November 2026, by researchers in robotic systems, smart grid security, and flying communication networks

Heterogeneous Generative Dataset for UASes — 2023
IEEE International Conference on Mobility, Operations, Services and Technologies (MOST 2023), pp. 229–230

- Authors: Dharmalingam, B., **Odat, I.**, Mukherjee, R., Piggott, B., & Liu, A. (Second Author) · doi:10.1109/MOST57249.2023.00034

- Contributed to dataset design and evaluation methodology for standardized benchmarking of adversarial attacks on unmanned aerial systems; addresses the absence of heterogeneous training data for UAS security evaluation

PREPRINTS & MANUSCRIPTS UNDER REVIEW

AeroMind: Poisoning the Control Plane of LLM-Driven UAV Agents 2026
Under review at ACM AsiaCCS 2027

- Authors: **Odat, I.**, Liu, A., & Li, Y. (First Author)
- Evaluated a multi-agent UAV system across 15 attack scenarios and seven planner backends, demonstrating that three poisoned memory records can redirect coordinated mission behavior and spread between agents through a shared-memory architecture
- Developed a three-layer defense pipeline (provenance, trust reranking, diversity cap) reducing measured attack success from 100% to 0% in controlled evaluation
- Proposed and formalized the first documented threat model for memory-poisoning attacks in LLM-powered multi-agent UAV systems

Beyond Belief: Evaluating and Mitigating LLM Susceptibility to Misinformation 2026
arXiv preprint arXiv:2610.33962

- Authors: Kanaan, E., Lee, J., **Odat, I.**, Vosoughi, S., & Ma, W.
- Tests whether the credibility judgments of tool-using LLMs predict whether they act on misinformation, across five models, 196 constructed false notices, and 204 archived attacks, with forewarning and fine-tuning as mitigations

RESEARCH EXPERIENCE

Graduate Research Assistant (GRA) Jan. 2022 – Present
Oakland University · Advisor: Dr. Anyi Liu Rochester, MI

- Designed and implemented AeroMind, a multi-agent UAV fleet integrating large language models (LLMs), persistent vector memory, and PX4 software-in-the-loop (SITL) flight simulation; serves as the primary research platform for the doctoral dissertation on memory-poisoning attacks and defenses in autonomous UAV systems
- Penetration-tested PX4 open-source drone flight controllers, identifying buffer overflows, session hijacking, and MAVLink (drone communication protocol) exploits across three distinct attack vectors; developed and validated mitigations incorporated into the lab's standardized UAV security evaluation framework
- Architected AI-driven threat detection pipelines using fine-tuned LLMs, prompt engineering, and Ollama, demonstrating a 60% reduction in manual threat triage time in controlled lab evaluation
- Simulated MITM and brute-force attacks against UAV encryption protocols and IDS/IPS configurations; findings documented in technical reports and incorporated into subsequent UAV security research within the lab

TEACHING EXPERIENCE

Teaching Assistant (TA) Jan. 2022 – Dec. 2024
Oakland University, Department of Computer Science & Engineering Rochester, MI

- Teaching Assistant for four graduate and undergraduate cybersecurity and systems courses, supporting instruction for both M.S. and B.S. level students
- **CSI 5480: Information Security Practices** · graduate-level instruction on security principles, threat modeling, and defensive techniques
- **CSI 6600: Network Security** · graduate-level coursework covering network threat analysis, intrusion detection, and secure communications
- **CSI 4500/5500: Fundamentals of Operating Systems** · OS architecture, process management, and system-level security concepts
- **CSI 1420: Introduction to C Programming and Unix** · undergraduate systems programming fundamentals

PRESENTATIONS & RESEARCH OUTREACH

Research Presentation, Engineering Day Jan. 24, 2026
Oakland University School of Engineering and Computer Science Rochester, MI

- Presented doctoral research on LLM-enabled autonomous drone security to prospective students, families, and faculty at the annual Engineering Day public outreach event

- Demonstrated the AeroMind multi-agent drone framework end to end, covering the autonomy pipeline, real-time control, AI perception and decision-making, and memory-based attacks and defenses across DJI Tello and PX4-based drone platforms

OPEN-SOURCE RESEARCH PROJECTS

AeroMind Drone Security Testbed github.com/OdatSec/AeroMind

- Multi-agent UAV research platform with a supervisor agent, autonomous scout drones, and persistent shared memory, built to evaluate memory poisoning, cross-agent propagation, and retrieval-stage defenses. Python, PX4 SITL, MAVSDK, Gazebo, Ollama

Net-GPT Evaluation Toolkit github.com/OdatSec/Net-GPT

- Reusable evaluation workflows for fine-tuning language models and generating counterfeit UAV network packets, released to enable independent replication of the SEC 2023 paper. Python, LLM fine-tuning, MAVLink

PX4 Autopilot Session Hijacking Exploit PX4, MAVLink, MAVSDK, Python

- Reverse-engineered MAVLink session management to achieve privilege escalation on PX4 Autopilot; built a proof-of-concept exploit demonstrating telemetry stream hijacking, with remediation proposals documented for the research community

TECHNICAL SKILLS

AI/LLM Security: Adversarial ML, LLM fine-tuning (QLoRA), prompt engineering, RAG poisoning, memory poisoning, LangGraph, LangChain, Ollama

Autonomous Systems: ROS 2, PX4 Autopilot, Gazebo, MAVSDK, MAVLink, NVIDIA Jetson Nano, YOLOv8

Offensive Security: Penetration testing, exploit development, firmware analysis, reverse engineering, red teaming, OSINT

Frameworks & Standards: MITRE ATT&CK, OWASP Top 10, NIST SP 800-171, NIST AI RMF, PCI DSS, ISO 27001

Security Tooling: Kali Linux, Metasploit, Burp Suite, Wireshark, Nessus, Nmap, SQLMap, SIEM, IDS/IPS

Programming & Infrastructure: Python, Bash, C/C++, SQL, FastAPI, Docker, AWS

CERTIFICATIONS & MEMBERSHIPS

Certified Ethical Hacker (CEH), EC-Council, 2023 · **eLearnSecurity Junior Penetration Tester (eJPT)**, INE, 2022 · **Fortinet NSE 1 & NSE 2**, Fortinet, 2021

Member, **IEEE** · Member, **ACM**

PRIOR INDUSTRY EXPERIENCE

Cybersecurity Specialist Jan. 2021 – Jan. 2022
ZADD Amman, Jordan

- Integrated AI-based anomaly detection into an enterprise SaaS platform, reducing false-positive security alerts by 35%; managed enterprise-wide NIST SP 800-171 compliance remediation across critical infrastructure

Penetration Tester Jan. 2020 – Jan. 2021
Futuretec Security Solutions Amman, Jordan

- Conducted systematic web application and API security assessments identifying critical injection, authentication, and configuration vulnerabilities; developed custom Python automation frameworks reducing manual testing effort by 50%